

Portable Hearing Test System Statement of Work

1. General Requirement

V Corps Headquarters and Headquarters Battalion (HHBn) requires two (2) portable hearing test systems with the following characteristics.

2. System Characteristics

- **2.1. Core Functionality:** The system shall provide a portable, automated solution for conducting hearing tests and hearing protection fit tests. It must be designed to replace the need for a traditional sound-proof booth, an external audiometer, and associated wires, combining these functions into a single, compact system.
- **2.2. Operation:** The headset shall connect wirelessly to a control unit (such as a tablet) that guides the user through a self-administered test. The system must be automated to ensure reliable and repeatable results, supported by scientific validation. The user interface must be intuitive, allowing operation by personnel without specialized technical skills. The system must also permit manual test administration by a qualified audiologist or technician.
- **2.3. Throughput:** The system must support the simultaneous testing of multiple individuals to allow for high-throughput screening. A single administrator must be able to monitor several simultaneous tests.
- **2.4. Data Management:** All test results must be exportable in both PDF and CSV formats. Data shall be stored securely on the local device until transferred by an administrator.
- **2.5. Test Duration:** The average time for a complete audiometric hearing test should be approximately 7-10 minutes. The hearing protection fit test should take an additional 3-5 minutes.

3. Technical Performance

- **3.1. Noise Attenuation:** The headset must provide high passive ambient noise attenuation, equivalent to or exceeding that of a single-wall sound booth. The required attenuation must be sufficient to conduct compliant hearing tests in environments with ambient noise levels up to at least 60 dBA. The vendor must provide independent laboratory data verifying the attenuation performance of the headset across the 250-8000 Hz frequency range.
- **3.2. Ambient Noise Tolerance:** The system must be capable of performing accurate audiometric testing in a variety of non-clinical environments, such as classrooms or conference rooms, without a sound booth. It must demonstrate

high maximum permissible ambient noise levels (MPANLs), particularly in the low frequencies (250-1000 Hz), as calculated per ANSI S3.1-1999 (R2018).

- **3.3. Validation:** The system's accuracy and reliability must be supported by independent, peer-reviewed scientific studies comparing its performance to current clinical standard audiometers in relevant environments. The vendor must provide citations to or copies of such studies.
- **3.4. Ambient Noise Monitoring:** The system should be compatible with a commercially available sound level meter to allow for continuous monitoring of ambient noise during testing.

4. Technology Readiness & Manufacturing

- **4.1. Technology Readiness:** The proposed system must be a commercially available, off-the-shelf (COTS) solution. It must have a proven track record of successful deployment and use in military, industrial, or clinical hearing conservation programs.
- **4.2. Manufacturing:** The system must be manufactured in the United States.

5. Hearing Protection Fit Test Capability

- **5.1. Compatibility:** The system's hearing protection fit test function must be compatible with a wide variety of in-ear hearing protection devices, including foam, flanged, and custom-molded earplugs.
- **5.2. Methodology:** The fit test must comply with the standards outlined in DoDI 6055.12. The system shall calculate a Personal Attenuation Rating (PAR) by comparing hearing thresholds with and without the hearing protection in place, using 1/3 octave noise stimuli at a minimum of 500, 1000, and 2000 Hz.

6. Data Integration and Security

- **6.1. DOEHRS Integration:** The system must be capable of integration with the Defense Occupational and Environmental Health Readiness System (DOEHRS). The vendor must describe their current integration status or provide a detailed plan and timeline for achieving direct, automated data transfer to DOEHRS-HC.
- **6.2. Cybersecurity:** The system must meet all applicable Department of Defense (DoD) and Defense Health Agency (DHA) cybersecurity requirements. The vendor must provide evidence of a current cybersecurity approval (e.g., an Assess & Incorporate package or Authority to Operate) under the Risk Management Framework (RMF). The system must ensure the security of Personally Identifiable Information (PII) and Protected Health Information (PHI) at all times, with data encrypted both at rest and in transit.

7. Calibration and Maintenance

- **7.1. Annual Calibration:** The system must be capable of being calibrated annually to meet all applicable DoD and ANSI standards. The vendor shall describe the required calibration process, including services offered.
- **7.2. Daily Verification:** The system must include a daily automated self-check or bio-acoustic simulation function to verify that it is functioning within calibration specifications prior to the start of testing.
- **7.3. Warranty:** The system shall include a comprehensive warranty. Vendors must specify the duration and terms of the warranty. A warranty period of five (5) years is desired.

8. Compliance and Portability

- **8.1. Regulatory Compliance:** The system must be manufactured in an ISO 13485-certified facility, be registered with the FDA, and be fully compliant with the following standards:
 - **Hearing Testing:** ANSI S3.6, OSHA 1910.95, IEC 60645-1
 - **Fit Testing:** ANSI S12.71, DoDI 6055.12
- **8.2. Portability:** The system must be designed for maximum portability and use in field environments. It must be delivered in a rugged, padded, protective case that is suitable for transport and sized to fit under a commercial aircraft seat.